10 Similarity Search Trees: Approximate Nearest Neighbors Search for Image Retrieval

This chapter covers

- Discussing the limits of k-d trees
- Describing image retrieval as a use case where k-d trees would struggle
- Introducing a new data structure, the R-tree
- Presenting SS-trees, a scalable variant of R-trees
- Comparing SS-trees and k-d trees
- Introducing approximate similarity search

This chapter will be structured slightly differently from our book's standard, simply because we will continue here a discussion started in chapter 8. Back there, we introduced a problem, searching multidimensional data for the nearest